

Generation Interconnection System Impact Study Report

For Queue Project Q15-0493

Capitol Cement 100 kV

September 02, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0493. The Interconnection Customer proposes a 188.9 MW battery energy storage facility interconnecting at the Capitol Cement 100 kV substation in the Illinois area. The request was received on June 26, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 176 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$32,379,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Bammel to Capitol Cement 100 kV is overloaded to 209.4% of its 116 MVA emergency rating for the loss of the Ocoee 2 to Mt Gate Junction 100 kV L. The project contributes 100.0% DFAX (188.9 MW) to this overload.

The Dayton'S Bluff to Bammel 100 kV is overloaded to 181.2% of its 120 MVA emergency rating for the loss of the Mt Gate Jct to Ocoee 2 100 kV. The project contributes 70.7% DFAX (133.5 MW) to this overload.

The Ocee 2 to Mt Gate Jct 100 kV is overloaded to 178.3% of its 137 MVA emergency rating for the loss of the Capitol Cement to Bammel 100 kV L. The project contributes 100.0% DFAX (188.9 MW) to this overload.

Ocoee 2 to Capitol Cement Substation 100 kV L: 167.5% loading (129 MVA) following the outage of the Capitol Cement to Bammel 100 kV; DFAX = 100.0%.

For the loss of Dayton'S Bluff to Bammel 100 kV line, the Cooper Nuclear Station to Storey 100 kV loads to 130.0% of its 131 MVA rating. Project DFAX: 47.0%.

For the loss of Dayton'S Bluff to Bammel 100 kV, the Lewis Smith Dam Substation to Storey 100 kV L loads to 116.8% of its 130 MVA rating. Project DFAX: 47.0%.

Tuscaloosa T.S to Cooper Nuclear Station 100 kV Circuit 1: 116.0% loading (144 MVA) following the outage of the Cooper Nuclear Sta to Tuscaloosa T.S 100 kV L; DFAX = 12.2%.

Tuscaloosa T.S to Cooper Nuclear Sta 100 kV Cir. 2: 116.0% loading (144 MVA) following the outage of the Cooper Nuclear Station to Tuscaloosa T.S 100 kV Circuit 1; DFAX = 12.2%.

The Cypress Creek to Big Oaks 100 kV is overloaded to 115.3% of its 241 MVA emergency rating for the loss of the Jones Chapel to Dekalb 500 kV. The project contributes 5.4% DFAX (10.2 MW) to this overload.

For the loss of Lost Canyon to Mountain Island 100 kV, the San German 6401 to Dayton'S Bluff 100 kV line loads to 110.3% of its 110 MVA rating. Project DFAX: 16.4%.

The Oakmot N to Mt Gate Jct 100 kV Ckt 2 is overloaded to 110.0% of its 160 MVA emergency rating for the loss of the Oakmont N to Mt Gate Junction 100 kV Line 1. The project contributes 22.6% DFAX (42.7 MW) to this overload.

For the loss of Bammel to Dayton'S Bluff 100 kV, the Seton to Lewis Smith Dam 100 kV line loads to 109.6% of its 106 MVA rating. Project DFAX: 47.0%.

Ligonier to Cypress Crk 100 kV: 107.1% loading (269 MVA) following the outage of the Jones Chapel to Dekalb 500 kV; DFAX = 5.4%.

The Tuscaloosa T.S to Texon 100 kV Circuit 1 is overloaded to 106.0% of its 179 MVA emergency rating for the loss of the Tuscaloosa T.S to Texon 100 kV Circuit 2. The project contributes 12.2% DFAX (23.1 MW) to this overload.

For the loss of Tuscaloosa T.S to Texon 100 kV Line 1, the Tuscaloosa T.S to Texon 100 kV line loads to 105.3% of its 180 MVA rating. Project DFAX: 12.2%.

Venus Substation to Oakmont N 100 kV Circuit 2: 100.5% loading (192 MVA) following the outage of the Oakmont North to Venus 100 kV Line 1; DFAX = 23.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Capitol Cement to Bammel 100 kV	Rebuild line with higher-capacity conductor	\$3,318,000	100% (\$3,318,000)
Dayton'S Bluff to Bammel 100 kV	Rebuild line with higher-capacity conductor	\$3,231,000	100% (\$3,231,000)
Mt Gat Junction to Ocoee 2 100 kV L	Rebuild line with higher-capacity conductor	\$8,473,000	100% (\$8,473,000)
Ocoee 2 to Capitol Cement 100 kV	Rebuild line with higher-capacity conductor	\$3,320,000	100% (\$3,320,000)
Cooper Nuclear Station to Storey 100 kV	Rebuild line with higher-capacity conductor	\$3,284,000	100% (\$3,284,000)
Lewis Smith Dam Substation to Storey 100 kV L	Rebuild line with higher-capacity conductor	\$1,317,000	100% (\$1,317,000)
Tuscaloosa T.S to Cooper Nuclear Station 100 kV Circuit 1	Rebuild line with higher-capacity conductor	\$1,209,000	88% (\$1,063,000)
Tuscaloosa T.S to Cooper Nuclear Sta 100 kV Cir. 2	Rebuild line with higher-capacity conductor	\$2,281,000	88% (\$2,005,000)
Big Oaks to Cypress Crk 100 kV	Rebuild line with higher-capacity conductor	\$7,679,000	4% (\$266,000)
San German 6401 to Dayton'S Bluff 100 kV line	Rebuild line with higher-capacity conductor	\$9,572,000	27% (\$2,610,000)
Mt Gate Junction to Oakmont N 100 kV Ckt 2	Reconductor line	\$783,000	100% (\$783,000)
Seton to Lewis Smith Dam 100 kV line	Reconductor line	\$1,302,000	100% (\$1,302,000)
Ligonier to Cypress Crk 100 kV	Reconductor line	\$3,219,000	4% (\$112,000)
Tuscaloosa T.S to Texon 100 kV	Reconductor line	\$338,000	88% (\$297,000)
Tuscaloosa T.S to Texon 100 kV	Reconductor line	\$552,000	88% (\$485,000)
Venus Substation to Oakmont N 100 kV Circuit 2	Reconductor line	\$513,000	100% (\$513,000)
	Total allocated to project		\$32,379,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.